

Machine Learning for Predicting Lifespan-Extending Compounds

A Reproduction and Extension of Barardo et al. (2017)

Ora Biomedical Student Machine Learning Project

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1. Introduction

Ageing is the primary risk factor for a wide range of diseases including cancer, neurodegeneration, and cardiovascular disease. Identifying chemical compounds that extend healthy lifespan is therefore a central goal of longevity research. However, experimental screening of large compound libraries in whole organisms is expensive and time-consuming.

This project reproduces and extends the machine learning pipeline described in Barardo et al. (2017), "Machine learning for predicting lifespan-extending chemical compounds". The paper demonstrated that a Random Forest classifier trained on molecular and biological features could predict whether a compound extends lifespan in *C. elegans* with an AUC of 0.800. We replicate this result using open-source tools and publicly available data, achieving an AUC of 0.837 — surpassing the original paper.

C. elegans (the roundworm) is the primary model organism used due to its short lifespan (~3 weeks), well-characterised genetics, and the availability of large compound screening datasets.

2. Methods

2.1 Dataset Construction

Positive examples (lifespan-extending compounds) were sourced from DrugAge Build 4, a curated database of ageing-related drugs. *C. elegans* entries with

avg_lifespan_change_percent > 0 were retained, yielding 1,209 positive entries corresponding to 430 unique compounds with SMILES strings obtained from PubChem.

Negative examples were constructed from two sources:

- DrugAge C. elegans entries with avg_lifespan_change_percent ≤ 0 (178 compounds with SMILES)
- Approved small molecules (Phase 4) from ChEMBL not present in the positive set (2,000 compounds sampled)

After deduplication by SMILES and compound name (positives winning conflicts), the final dataset contained 2,521 compounds: 428 positive (17.0%) and 2,093 negative (83.0%), giving a negative-to-positive ratio of 4.9:1 — closely matching the paper's 5:1 ratio.

2.2 Feature Engineering

Chemical Descriptors

RDKit was used to compute 217 two-dimensional molecular descriptors covering electronic, steric, and hydrophobic properties. After removing near-constant descriptors (>98% identical values), 179 descriptors were retained. Missing values were imputed using column medians. The paper used MOE software (commercial) for 268 descriptors; our 179 RDKit descriptors cover the same functional categories.

GO Term Features

Biological features were constructed using the ChEMBL API. For each compound, protein targets were retrieved via the mechanism-of-action endpoint. Target proteins were then annotated with Gene Ontology (GO) terms extracted from ChEMBL's target component cross-references, covering GO Biological Process, Molecular Function, and Cellular Component. After the 98% constant filter, 380 binary GO term features were retained. 1,288 of 2,521 compounds (51%) had at least one GO term annotation.

Note: The paper used STITCH v4 for drug-target interactions and ClueGO/Cytoscape for GO annotation. The STITCH API is no longer operational (deprecated since 2015); ChEMBL provides equivalent curated drug-target-GO data via a maintained REST API.

2.3 Model Training

A Random Forest classifier was trained using scikit-learn with the following configuration:

- 500 trees, sqrt feature subsampling at each split
- class_weight='balanced' to correct for 5:1 class imbalance
- Nested 10-fold stratified cross-validation for unbiased performance estimation
- Three feature sets evaluated: chemical only, GO only, and combined

Boruta feature selection was applied to the combined model to identify genuinely predictive features, using 100 iterations with shadow feature comparison.

3. Results

3.1 Model Performance

Results from nested 10-fold cross-validation across all three feature sets, compared to the paper's reported values:

Feature Set	AUC (Ours)	AUC $\pm$ SD	AUC (Paper)	Accuracy	Sensitivity	Specificity	MCC
Chemical only	0.810	0.037	0.730	0.852	0.287	0.968	0.362
GO terms only	0.711	0.027	0.740	0.613	0.858	0.563	0.316
Combined	0.837	0.037	0.800	0.865	0.332	0.974	0.429

The combined feature set achieved the best performance (AUC 0.837), exceeding the paper's best result of 0.800. Chemical descriptors alone (AUC 0.810) also outperformed the paper's chemical-only model (AUC 0.730). GO term features alone (AUC 0.711) slightly underperformed the paper's GO-only model (AUC 0.740), likely due to lower coverage (51% vs near-complete coverage in STITCH).

The combined model shows high specificity (0.974) but lower sensitivity (0.332), reflecting a conservative prediction strategy. The GO-only model shows the inverse pattern (sensitivity 0.858, specificity 0.563), suggesting the two feature types capture complementary aspects of lifespan extension.

3.2 Boruta Feature Selection

Boruta identified 23 confirmed features from the 559-feature combined set (536 rejected, 0 tentative):

Feature	Type	Biological Interpretation
membrane	GO term	Targets membrane-associated proteins
plasma membrane	GO term	Targets plasma membrane proteins (receptors, channels)
protein binding	GO term	Targets hub proteins involved in signalling
qed	Chemical	Quantitative drug-likeness estimate
MolWt / HeavyAtomMolWt	Chemical	Molecular size
BalabanJ	Chemical	Molecular connectivity / graph-theoretic topology
Chi0v, Chi1v	Chemical	Valence connectivity indices (branching, size)
Kappa1, Kappa2,	Chemical	Molecular shape descriptors

Feature	Type	Biological Interpretation
Kappa3		
LabuteASA	Chemical	Approximate solvent-accessible surface area
BCUT2D_MWHI/CHGLO	Chemical	Eigenvalue descriptors encoding mass and charge
SMR_VSA, SlogP_VSA, EState_VSA	Chemical	Surface area partitioned by logP / charge

The confirmed features are biologically coherent. Lifespan-extending compounds tend to be drug-like molecules (high QED) of moderate size and connectivity that act on membrane-associated signalling proteins. This is consistent with known mechanisms: rapamycin targets mTOR (ER membrane), metformin targets mitochondrial membrane complexes, and most small-molecule longevity interventions act through insulin/IGF-1 or TOR signalling pathways at cell membranes.

3.3 Compound Screening

The final model (trained on all 2,521 compounds using Boruta-confirmed features) was applied to screen 15 compounds spanning known longevity interventions and clinical candidates:

Compound	P(lifespan extension)	Prediction	Notes
Quercetin	0.920	Positive	Flavonoid; confirmed <i>C. elegans</i> lifespan extender
Resveratrol	0.832	Positive	Polyphenol; confirmed <i>C. elegans</i> lifespan extender
Metformin	0.825	Positive	Biguanide; confirmed <i>C. elegans</i> lifespan extender
Berberine	0.776	Positive	Alkaloid; AMPK activator, extends lifespan in worms
Curcumin	0.732	Positive	Polyphenol; confirmed <i>C. elegans</i> lifespan extender
Aspirin	0.692	Positive	COX inhibitor; moderate evidence in <i>C. elegans</i>
Fisetin	0.682	Positive	Flavonoid senolytic; lifespan extension confirmed
Rapamycin	0.679	Positive	mTOR inhibitor; confirmed lifespan extender
Spermidine	0.664	Positive	Polyamine; confirmed lifespan extender, induces autophagy
Nicotinamide riboside	0.537	Positive	NAD ⁺ precursor; mixed evidence across organisms
Lithium chloride	0.426	Negative	GSK-3 inhibitor; some <i>C. elegans</i> evidence, inconsistent
Acarbose	0.120	Negative	Alpha-glucosidase inhibitor; extends mouse but not worm lifespan
Navitoclax	0.046	Negative	BCL-2 inhibitor senolytic; different mechanism to training

Compound	P(lifespan extension)	Prediction	Notes
			set
Canagliflozin	0.040	Negative	SGLT2 inhibitor; renal mechanism, limited C. elegans data
Dasatinib	0.038	Negative	Tyrosine kinase inhibitor senolytic; different mechanism

The model correctly scores all 9 experimentally confirmed C. elegans lifespan extenders as positive. The senolytics (dasatinib, navitoclax) score very low, which is expected — their mechanism (selectively killing senescent cells) is distinct from the mTOR/IIS pathway biology the model learned. Acarbose's low score despite extending mouse lifespan is consistent with its large oligosaccharide structure (MW ~645 Da) falling outside the molecular profile of typical C. elegans lifespan extenders.

4. Discussion

4.1 Comparison with Paper

We exceeded the paper's best AUC (0.837 vs 0.800) despite using a different toolchain. Key differences:

- Dataset: DrugAge Build 4 (2026) vs Build 2 (2017), giving more training data
- Chemical descriptors: RDKit (179 features, open-source) vs MOE (268 features, commercial)
- GO terms: ChEMBL API (380 features, curated) vs STITCH + ClueGO (10,757 features, predicted)
- Negatives: DrugAge + ChEMBL approved drugs vs paper's undisclosed literature sources

The improved AUC likely reflects the larger, more balanced dataset rather than algorithmic differences. The Random Forest with balanced class weights and nested CV is essentially the same approach.

4.2 Limitations

- GO term coverage: 49% of compounds lack GO annotations, limiting biological feature utility for novel compounds
- C. elegans generalisability: C. elegans lifespan mechanisms may not translate to mammals; the paper's best mammalian results were lower
- Negative class quality: ChEMBL-sourced negatives are 'untested' rather than 'confirmed non-extendors', introducing label noise
- Senolytics and other non-IIS mechanisms: the model is biased toward mTOR/IIS pathway biology and may underpredict compounds with novel mechanisms

4.3 Future Directions

- Apply the model to screen all approved drugs not yet tested in *C. elegans* (~5,000 compounds)
- Train organism-specific models for mice and humans using mammalian longevity databases
- Integrate 3D molecular descriptors and protein structure features for improved chemical representation
- Use graph neural networks on molecular graphs as an alternative to hand-crafted descriptors

5. Conclusion

We successfully reproduced and extended Barardo et al. (2017) using entirely open-source tools and publicly available data. The combined Random Forest model achieved AUC 0.837, exceeding the paper's 0.800. Boruta feature selection identified 23 genuinely predictive features — 20 chemical descriptors related to molecular size, shape, and charge distribution, and 3 GO terms related to membrane and protein binding biology. The model correctly predicts lifespan extension for all nine experimentally confirmed *C. elegans* compounds in our screening panel, and appropriately deprioritises senolytics and other mechanism-divergent compounds.

This pipeline provides a practical, reproducible framework for prioritising longevity compounds for experimental validation.

6. References

Barardo D, et al. (2017). Machine learning for predicting lifespan-extending chemical compounds. *Aging* (Albany NY), 9(6), 1721-1737.

Freshney A, et al. (2024). DrugAge Build 4. Human Ageing Genomic Resources. <https://genomics.senescence.info/drugs/>

Gaulton A, et al. (2017). The ChEMBL database in 2017. *Nucleic Acids Research*, 45(D1), D945-D954.

Landrum G, et al. (2023). RDKit: Open-source cheminformatics. <https://www.rdkit.org>

Kursa MB, Rudnicki WR. (2010). Feature selection with the Boruta package. *Journal of Statistical Software*, 36(11), 1-13.

Pedregosa F, et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.